

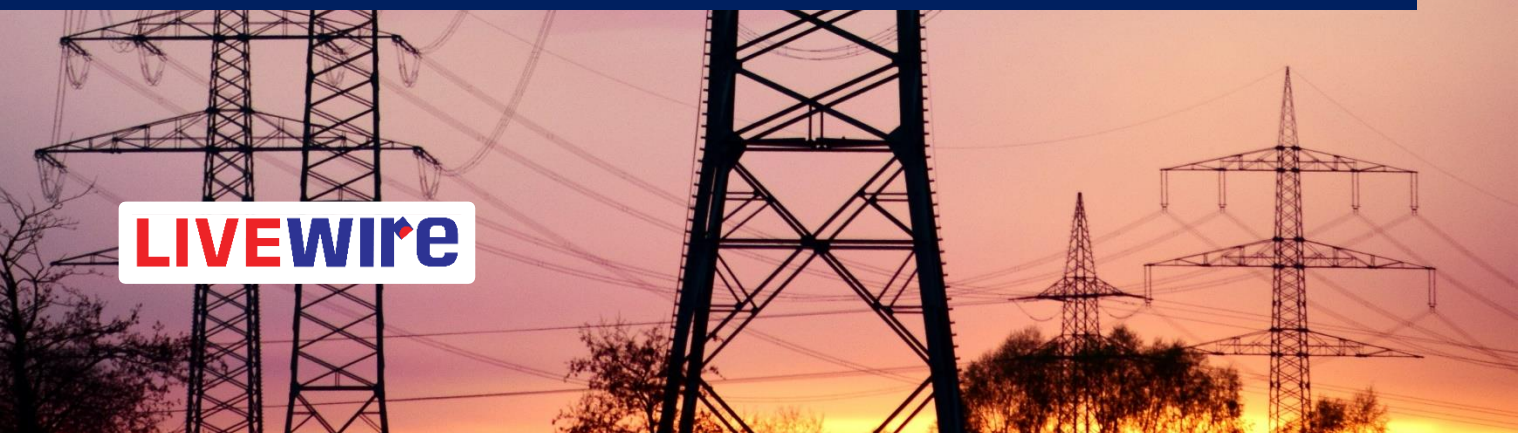
Electrical Power System Analysis

Day Wise Curriculum

LIVEWIRE - A Division Of CADD Centre Training Services Pvt Ltd

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Electrical Power System Analysis

Day 1

Introduction to Electrical System Design
Application of Electrical Design
Purpose of Electrical Design
Basic Design philosophy
Introduction to Electrical Power system
Structure of Electrical power system

Day 2

Sources of Electrical Energy
- Conventional Energy sources
- Non- Conventional Energy sources

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Day 3

Elements of Power system
- Generators
- Transformers
- Transmission Line
- Bus bars
- Isolators
- Lightning Arresters
- Reactors
- Capacitors
- Wave Trap
- Grounding

Day 4

Generation, Transmission and Distribution
- Generation Transmission & Distribution systems
- Generating stations
- Types of Transmission & Distribution systems
- Types of systems for power transmission
- Basic Principle of Power System

Day 5

Introduction to Power Substation
- Substation
- Types of Substations based on the purpose.
- Types of Substations based on physical features.
- Substation layout

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Day 6

- Introduction to Bus bar
- Bus-Bar arrangements
- Bus-bar materials
- Substation bus schemes
- Substation location

Day 7

- Introduction to Single Line diagram
- Single Line diagram (or) One-line diagram
- SLD of Power System
- Importance of using Single line diagram
- Standard symbols in Single line diagram

Day 8

Constructing a Single line diagram

Day 9

Single Line Diagram Examples
Single Line diagram of 11kV/400V Substation
Single Line diagram of 66/11 kV Substation
Single Line diagram of 33/11kV Substation
Single Line diagram of 400/132/66/33kV Substation

Day 10

Design a Single Line Diagram

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Day 11

Introduction to ETAP
What is ETAP?
Why ETAP for System design
History of ETAP
Key features & Benefits of ETAP
Codes & Standards

Day 12

Working with ETAP software
- Starting ETAP software
- Creating a new project
- Changing the Project standard
- File Management
- Exploring GUI

Day 13

- Toolbar Description
  - Project Toolbar
  - Theme Toolbar
  - System Toolbar
  - Mode Toolbar
  - Base & Revision Toolbar

Day 14

- Inserting Circuit Elements
- Library for Circuit Elements
- System Elements and Components
- Element Classification
  - AC Elements
  - DC Elements
  - AC-DC Elements
  - Instrumentation Elements
- Component Editor

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Day 15

- Creating Single Line Diagram in ETAP

Day 16

- Editing a one-line diagram
- Add Elements
- Add Multiple Elements
- Connect Elements
- Select & Deselect Elements

Auto Select
Element & Connector Weight
Relocate elements.
Cut, copy, paste.
Group & Ungroup
Nodes & Buses

Day 17

Editing a one-line diagram
- Element Size
- Element symbols
- Element & Connector Colors
- Element Alignment
- Element Classification
- Element Rotation
- Get Template and Add to Template
- Connector Routing
- Element Status

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Day 18

Design a Single line diagram.

Day 19

Modules in ETAP
- Load Flow Analysis
- Short Circuit Analysis
- Motor Acceleration
- Transformer Sizing
- Cable Sizing
- Relay Co-ordination
- Harmonic Analysis

Arc Flash
Ground Grid System

Day 20

Module – 1 Load Flow Analysis

Day 21

Design a Single line diagram and run the Load Flow Analysis

Day 22

Module – 2 Short circuit analysis

Day 23

Design a Single line diagram and run the Short Circuit Analysis

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Day 24

Module – 3 Motor Acceleration

Day 25

Design a Single line diagram and run the Motor Acceleration

Day 26

Module – 4 Transformer Sizing

Day 27

Design a Single line diagram and run the Transformer Sizing

Day 28

- Introduction to Cables
  - Cables
  - Comparison of Underground cables and Overhead lines
  - General construction of cables
  - Type of Cables
  - Insulating materials for cables
- Module – 5 Cable Sizing

Day 29

- Introduction to Power System Protection
  - Power system protection
  - Need of protection in Power System
  - Protective system Elements
    - Fuse
    - Circuit breaker
    - Current Transformer
    - Potential transformer
- Types of Circuit breaker

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Day 30

- Types of Relays
  - Electromechanical Relay
  - Static Relay
  - Numerical Relay
  - Digital Relay
- Comparison of Different types of Relays
- Protective Relays
  - Electromagnetic attraction type relay
  - Induction type relay
  - Directional type relay
  - Distance type relay.
  - Differential type relay
- Module – 6 Relay Co-ordination

Design a Single line diagram and run the Relay Co-ordination

Day 31

Module – 7 Harmonics Analysis
Design a Single line diagram and run the harmonic.

Day 32

Module – 8 Arc Flash
Design a Single line diagram, run the Arc Flash, and check the output with standard values.

Day 33

Earthing system
- What is earthing?
- Why is earthing necessary?
- Types of earthing
- What factors affect earthing?
- Step potential.
- Touch potential.

Module – 9 Ground grid studies
- IEEE method
- Finite Element Method (FEM)

Day 34

Design a Single line diagram and run all the above-mentioned modules.

Day 35

Pre-Placement Training

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Day 36

Final Project Guidance and Discussion Forums

Day 37

Interview Preparations

Day 38

Final Project Guidance and Discussion

Day 39

Final Project submission and evaluation

Day 40

Final Project Report Submission

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